

Rigid Motions and Constructions — Planning and Justifying Compass-and-Straightedge Work

Answer Key

High School Geometry

Quick Answers

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|---------------|-------|------------|-------|
| 1. — | 4. — | 7. 10, 10 | 10. — |
| 2. — | 5. 10 | 8. — | |
| 3. 7.07, 7.07 | 6. — | 9. 5, 5, 5 | |
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Common wrong answers

3. *If they got 8: added the horizontal and vertical differences instead of squaring, adding, and taking the square root*
9. *If they got 7.21: measured from the midpoint of the horizontal side instead of from the constructed circumcentre*
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How to grade this sheet. Six of the ten problems have no single correct answer, only a correct *plan* or *argument*, and each one below lists what must be present for full credit. The recurring gap is a plan whose arcs have no stated centre or radius, and a justification that appeals to the picture instead of to equal radii. The four coordinate problems are the antidote: they hand back the output of a construction and ask the student to test the very property the construction promised.

1. Plan: perpendicular bisector of AB .**Full-credit plan:**

- Open the compass to any radius greater than half of AB . Place the point on A and draw arcs above and below the segment.
- *Without changing the opening*, place the point on B and draw arcs above and below, crossing the first two. Call the crossings M and N .
- Lay the straightedge on M and N and draw the line. That line is the perpendicular bisector of AB .

The condition on the opening: the radius must be more than half of AB . If it is exactly half, the arcs touch at one point (the midpoint) and give no second point to draw a line through; if it is less than half, the arcs never meet and there is nothing to join.

Grading: step 2 must say the opening is unchanged, and the crossing points must be named on *both* sides. A plan that draws one arc on each side gets only one crossing and cannot determine a line.

Open plan — open response.

2. Plan: bisector of angle DEF .**Full-credit plan:**

- Put the compass point on the vertex E and draw an arc of any radius crossing both rays. Call the crossings P (on ED) and Q (on EF).
- Put the compass point on P and draw an arc inside the angle. *Keeping the same opening*, put the point on Q and draw a second arc crossing it. Call the crossing R .
- Draw ray ER with the straightedge. It bisects the angle.

Which arcs must share an opening: the two in step 2, from P and from Q . That shared radius is what makes $PR = QR$; together with $EP = EQ$ (both drawn with the step-1 radius) and the shared side ER , it gives $\triangle EPR \cong \triangle EQR$ by SSS, hence $\angle PER \cong \angle QER$. Different openings in step 2 destroy $PR = QR$, the triangles are no longer congruent, and ER lands off the bisector.

Open plan — open response.

3. Coordinate check: the perpendicular bisector.

Property being tested: every point on the perpendicular bisector of AB is equidistant from A and B . So the test is $PA = PB$.

$$PA = \sqrt{(-1 - (-2))^2 + (8 - 1)^2} = \sqrt{1 + 49} = \sqrt{50}$$

$$PB = \sqrt{(-1 - 6)^2 + (8 - 7)^2} = \sqrt{49 + 1} = \sqrt{50}$$

Both equal $5\sqrt{2}$, which to the nearest hundredth is 7.07. The two distances agree exactly, so P passes the test and the construction is confirmed at this point.

Note the two radicands are the same numbers in the opposite order — the horizontal and vertical gaps swap between the two distances. That symmetry is what equidistance looks like in coordinates.

Watch for: $1 + 7 = 8$, adding the coordinate differences instead of squaring them, adding, and taking the square root.

4. Justify: why the perpendicular-bisector construction works.

(a) **Congruent triangles.** All four arcs were drawn with one compass opening r , so by construction

$$AM = BM = AN = BN = r$$

In $\triangle AMN$ and $\triangle BMN$: $AM = BM$, $AN = BN$, and MN is common. So $\triangle AMN \cong \triangle BMN$ by **SSS**.

(b) **From the congruence to the conclusion.** CPCTC gives $\angle AMN \cong \angle BMN$. Now compare $\triangle AMX$ and $\triangle BMX$, where X is where MN crosses AB : $AM = BM$, the included angles at M are the congruent ones just obtained, and MX is common, so those triangles are congruent by **SAS**. CPCTC then gives $AX = BX$ — so X is the midpoint — and $\angle AXM \cong \angle BXM$. Those two angles are a linear pair, so each is a right angle and $MN \perp AB$.

Grading: both halves need a named criterion. An answer that stops at “the arcs are equal so it must be the bisector” has asserted the conclusion, not proved it.

Open justification — open response.

5. Coordinate check: the inscribed hexagon.

The claim the construction depends on: in a regular hexagon inscribed in a circle, each side is exactly as long as the radius — which is why walking one unchanged compass opening around the circle closes up in six steps.

$$\text{side} = \sqrt{(5 - 10)^2 + (8.66 - 0)^2} = \sqrt{25 + 75.0} \approx \sqrt{100}$$

The side comes out to 10, matching the radius to the accuracy of the rounded coordinate. The construction's promise holds.

Why the hexagon and not some other polygon: the centre and two consecutive vertices form a triangle with two sides equal to the radius, and the side equal to the radius makes it equilateral, so its central angle is 60° — and six of those fill the full turn exactly.

6. Plan: a parallel through P by copying an angle.

Full-credit plan:

- Draw any line through P that also crosses ℓ ; call the crossing T . This is the transversal.
- Put the compass point on T and draw an arc crossing both ℓ and the transversal. Keeping the same opening, draw the matching arc centred at P crossing the transversal.
- Set the compass to the distance between the two crossing points of the arc at T , then swing that distance from where the arc at P meets the transversal, marking a point S .
- Draw line PS . It is parallel to ℓ .

The step that creates corresponding angles: steps 2 and 3 together, which copy the angle at T to P on the same side of the transversal. Because the copied angle and the original are corresponding angles and are congruent, the converse of the corresponding-angles postulate gives $PS \parallel \ell$.

Why not a protractor: a construction is a proof carried out with tools whose guarantees are exact — an unmarked straightedge and equal compass radii. A protractor reads an angle to about a degree, so the resulting line would be approximately parallel, and no theorem would apply to it.

Open plan — open response.

7. Coordinate check: copying a segment.

Property being tested: a copied segment is congruent to the original, so $CD = AB$.

$$AB = \sqrt{(7 - 1)^2 + (10 - 2)^2} = \sqrt{36 + 64} = \sqrt{100}$$

$$CD = \sqrt{(11 - 3)^2 + (2 - (-4))^2} = \sqrt{64 + 36} = \sqrt{100}$$

Both are 10, so the copy is exact.

The two legs are 6 and 8 for AB and 8 and 6 for CD : the copy sits at a different angle, which is precisely the point — a compass transfers *length*, not direction.

8. Justify: the equilateral triangle.

Let r be the compass opening, set to AB , so $r = AB$.

The arc centred at A is part of the circle of radius r about A , and C lies on it, so $AC = r$ by the definition of a circle: every point of a circle is the same distance from the centre. The arc centred at B was drawn with the same opening, and C lies on it too, so $BC = r$ for the same reason.

Therefore

$$AC = r = AB \quad \text{and} \quad BC = r = AB$$

so $AB = AC = BC$ and $\triangle ABC$ is equilateral by definition.

Grading: the phrase that earns the credit is “all radii of a circle are equal” (or the definition of a circle). An answer that says the three sides “look the same length” or that measures them has not used a construction argument at all.

Open justification — open response.

9. Coordinate check: the circumcentre.

What two perpendicular bisectors give: a point equidistant from all three vertices — the circumcentre, the centre of the circle through A , B and C . So the test is $OA = OB = OC$.

$$OA = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25}$$

$$OB = \sqrt{(4 - 8)^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25}$$

$$OC = \sqrt{4^2 + (3 - 6)^2} = \sqrt{16 + 9} = \sqrt{25}$$

All three come out to 5, so O is the circumcentre and the circumradius is 5.

A useful check on the check: the triangle has its right angle at $A(0, 0)$, and for a right triangle the circumcentre is the midpoint of the hypotenuse. The midpoint of BC is $\left(\frac{8+0}{2}, \frac{0+6}{2}\right) = (4, 3)$, exactly the constructed point.

Watch for: measuring from the midpoint of AB , the point $(4, 0)$, which gives $\sqrt{4^2 + 6^2} \approx 7.21$ to C and unequal distances — the symptom of using one bisector instead of two.

10. Challenge: the quarter point, and why trisection is different.(a) **Plan.**

- Construct the perpendicular bisector of AB (problem 1's plan). It meets AB at the midpoint M .
- Construct the perpendicular bisector of AM the same way. It meets AM at Q .
- Q is the required point: AQ is one quarter of AB .

(b) **Justification.** Each bisection is justified exactly as in problem 4: the four equal arc radii give congruent triangles by SSS, CPCTC gives the equal halves, so the crossing point really is a midpoint. Applying it twice,

$$AM = \frac{1}{2}AB \quad \text{and} \quad AQ = \frac{1}{2}AM = \frac{1}{4}AB$$

The construction is exact, not approximate, because every step rests on equal radii rather than on a measurement.

(c) **Why repeated bisection cannot trisect.** Halving repeatedly can only ever produce points at $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{8}$ and so on — fractions whose denominators are powers of two. One third is not such a fraction, so no number of bisections lands on it; the points get close but never arrive.

Worth adding: a *segment* can nevertheless be trisected with compass and straightedge, using parallel lines and similar triangles rather than bisection. An *angle* cannot be trisected in general by these tools at all — a much deeper result, and a good example of the difference between “my method fails” and “no method exists”.

Open synthesis — open response.